

JKBOSE CLASS 10TH - MATHEMATICS

Time: 3 Hours

Series: 10th ARF(SZ) JKUT 2024-25 - Series A-3-X | Code: 103-X | Total Questions: 40
Max Marks: 80 Marks**GENERAL INSTRUCTIONS:** Section A (1 mark each), Section B (2 marks each), Section C (3 marks each), Section D (4 marks each)**SECTION - A : OBJECTIVE & VERY SHORT QUESTIONS**

(1 Mark Each)

1. Find zeroes of the polynomial $4u^2 + 8u$. [1 Mark]
2. Define an impossible event. [1 Mark]
3. 0.1 can be the probability of an event. (State True or False) [1 Mark]
4. How many tangents can a circle have? [1 Mark]
5. Circumference of a circle with radius 'r' = [1 Mark]
6. $\sqrt{3}$ is a rational number. (State True or False) [1 Mark]
7. The sum of a rational and an irrational number is always: [1 Mark]
 - (A) A rational number
 - (B) A natural number
 - (C) An irrational number
 - (D) An integer
8. HCF of (12, 15, 21) is [1 Mark]
9. The graph of $y = 4x$ is a line: [1 Mark]
 - (A) Parallel to X-axis
 - (B) Parallel to Y-axis
 - (C) Passing through origin
 - (D) None of these
10. If the pair of linear equations has a unique solution, then the lines representing these equations: [1 Mark]
 - (A) Coincide
 - (B) Intersect at one point
 - (C) Are parallel to each other
 - (D) None of these
11. The 30th term of the A.P. : 10, 7, 4, ... is: [1 Mark]
 - (A) 97
 - (B) -77
 - (C) 77
 - (D) -87
12. The value of $\cos \theta$ increases as θ increases. (State True or False) [1 Mark]
13. Evaluate: $(2 \tan 30^\circ) / (1 - \tan^2 30^\circ)$ [1 Mark]

14. If $an = n^2 + 1$, find a_3 . [1 Mark]
15. A quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) has two distinct real roots if: [1 Mark]
(A) $D = 0$
(B) $D < 0$
(C) $D > 0$
(D) None of these
16. Find the sum of the first five multiples of 3. [1 Mark]
17. Find the distance between the points (2, 3) and (4, 1). [1 Mark]
18. The mean of the first 'n' natural numbers is given by: [1 Mark]
(A) $n(n + 1) / 2$
(B) $n / 2$
(C) $(n^2 + 1) / 2$
(D) $(n + 1) / 2$
19. Find the mode of the given data: 3, 5, 7, 6, 1, 3, 2, 4, 3, 4. [1 Mark]
20. The volume of a sphere of radius 'r' is: [1 Mark]
(A) $\pi * r * l$
(B) $\pi * r^2 * h$
(C) $(1/3) \pi * r^2 * h$
(D) $(4/3) \pi * r^3$

SECTION - B : SHORT ANSWER QUESTIONS

(2 Marks Each)

21. Find whether the pair of linear equations $2x - 3y = 8$ and $4x - 6y = 9$ are consistent or inconsistent. [2 Marks]
22. Find the discriminant of the quadratic equation $2x^2 - 6x + 3 = 0$ and hence state the nature of its roots. [2 Marks]
23. If $\sin A = 3/4$, find $\cos A$ and $\tan A$. [2 Marks]
24. Find the point on the X-axis which is equidistant from (2, -5) and (-2, 9). [2 Marks]

--- OR ---

Find the value of 'y' for which the distance between the points P(2, -3) and Q(10, y) is 10 units.

25. Find a quadratic polynomial whose sum and product of zeroes are respectively 1 and 1. [2 Marks]
26. Find the area of the sector of a circle with radius 6 cm if the angle of the sector is 60 deg. [2 Marks]

SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

27. Find the area of a rhombus whose vertices are (3, 0), (4, 5), (-1, 4) and (-2, -1) taken in order. [3 Marks]

28. Prove that a parallelogram circumscribing a circle is a rhombus. [3 Marks]

--- OR ---

Prove that in two concentric circles, the chord of the larger circle which touches the smaller circle is bisected at the point of contact.

29. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes. [3 Marks]

30. Show that $3\sqrt{2}$ is an irrational number. [3 Marks]

31. If a line divides any two sides of a triangle in the same ratio, then prove that the line is parallel to the third side. [3 Marks]

--- OR ---

ABCD is a trapezium in which $AB \parallel DC$ and its diagonals intersect each other at point O. Show that $AO / BO = CO / DO$.

32. Find the sum of all odd numbers between 0 and 50. [3 Marks]

33. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting: (i) a spade, (ii) a red face card, (iii) a jack. [3 Marks]

34. D is a point on side BC of a triangle ABC such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB \cdot CD$. [3 Marks]

--- OR ---

E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that Triangle ABE ~ Triangle CFB.

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. Find the value of K so that the quadratic equation $Kx(x - 2) + 6 = 0$ has two equal real roots. [4 Marks]

--- OR ---

The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

36. A spherical glass vessel has a cylindrical neck 8 cm long and 2 cm in diameter; the diameter of the spherical part is 8.5 cm. Find the volume of the vessel. [4 Marks]

--- OR ---

A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its total surface area.

37. The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 60° . Find the height of the tower. [4 Marks]

--- OR ---

The angle of elevation of the top of a building from the foot of the

tower is 30 deg and the angle of elevation of the top of the tower from the foot of the building is 60 deg. If the tower is 50 m high, find the height of the building.

38. Prove the trigonometric identity: $[\cos A / (1 + \sin A)] + [(1 + \sin A) / \cos A] = 2 \sec A$ [4 Marks]

39. If AD and PM are medians of triangles ABC and PQR respectively, where Triangle ABC ~ Triangle PQR, prove that $AB / PQ = AD / PM$. [4 Marks]

--- OR ---

Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2 * \angle OPQ$.

40. The distribution below gives the weights of 30 students of a class. Find the median weight of the students: [4 Marks]

Weight (in kg)	Number of Students
40 - 45	2
45 - 50	3
50 - 55	8
55 - 60	6
60 - 65	6
65 - 70	3
70 - 75	2
Total	30